

Experiment Details

1 GPT Model Versions in Experiments

In our experiments, we used the OpenAI GPT-4o model for both our proposed framework and all baseline methods, serving as the backbone for tasks such as system prompt crafting (CraftSystemPrompt) and sample generation (GenerateSamples). Specifically, the Big Five and MobLab datasets were processed with gpt-4o-2024-05-13, while the World Values Survey dataset utilized gpt-4o-2024-08-06, following the deprecation of the earlier version at the time of the experiments.

2 Implementation Details

For our EM-based method, we set the number of prompts in the mixture model—denoted as K in Algorithm ??—based on the range of the target distribution for each game or dimension. We set $K = 10$ for all dimensions in the Big-Five dataset and for the Public Goods game in the MobLab dataset, and $K = 50$ for all other games in the MobLab dataset. We use the Wasserstein distance as the loss function and employ Sequential Least Squares Programming (SLSQP) to optimize the weights assigned to each prompt. For the optimization procedure, we initialize the weights randomly and set the tolerance parameter tol to $1e-6$. The maximum number of EM iterations is set to 20 for both datasets, with early stopping triggered if the selected prompts remain unchanged across iterations. When calling the GPT API, we use default parameter settings for all fields except model and messages.

For our GB-based method, we fix the maximum number of system prompts, K , to 200 for all tasks in both datasets. We adopt the same loss function (Wasserstein distance) and optimization procedure (SLSQP) as used in the EM-based method. The GPT API is also called in the same way, with default parameter settings except for the model and messages fields.

3 Baselines

We compare our approach against five baselines, covering fine-tuned, prompt-based, and human-derived methods. The fine-tuned LLaMA-3.1-model and OASIS are supervised baselines trained directly on our datasets. Persona Hub and Modular Pluralism are prompt-based methods not trained on our data. We also include a single fixed prompt baseline as a lower bound, and a distribution derived from 1,000 human responses as an upper bound.

For the fine-tuned Llama model, following the training paradigm of Be.FM, a foundation model for human behaviors [4], we take LLaMA-3.1-70B as the backbone model, fine-tune it on the MobLab and Big Five datasets separately. The model is fine-tuned models using LlamaFactory [7] with supervised learning and applied Low-Rank Adaptation (LoRA) [2] to all layers for memory-efficient training. For fine-tuning on the MobLab dataset, each human-play record is formatted as a data entry with the game instruction as the input and the observed behavior as the output, resulting in 82,057 training examples. For fine-tuning on the Big Five dataset, 17,667 subjects are used for training and 1,963 for evaluation. Two modeling tasks are performed: (1) Personality traits prediction, where each entry uses demographic information to predict one of the Big

Five scores; and (2) Demographics prediction, where each entry uses personality scores as input to predict the subject’s age. We then use the fine-tuned models as a generator across both datasets (generating Big Five survey question answers given demographics, and generating behaviors given MobLab game instructions), evaluating its capacity to approximate observed human distributions.

We use a single fixed system prompt as a lower-bound baseline to assess model (GPT-4o-2024-05-13/GPT-4o-2024-08-06) We use “*Imagine you are a human taking a personality test*” for Big-Five dataset, “*You are a helpful assistant.*” for MobLab dataset, and “*Imagine you are a human participant taking a values test.*” for World Values Survey dataset. This setting reflects a generic model response without perspective variation and serves as a contrast to more diversified methods.

Persona Hub [5] is a persona-driven data synthesis framework that leverages diverse internal perspectives within a large language model to generate high-variance synthetic data. It includes a large-scale collection of 1 billion personas automatically curated from web data, each acting as a distributed carrier of world knowledge. In our experiments, we evaluate the effectiveness of Persona Hub using the ElitePersonas dataset, which contains 370 million elite personas. Due to the scale of the dataset, we uniformly select one subset at random—subset 19, which contains 10,001,710 entries—as our sampling pool. From this subset, we uniformly sample 1,000 persona prompts without replacement and use them as system prompts for GPT-4o model.

OASIS [6] is an agent-based social media simulator designed to study large-scale group behaviors in digital environments. It can simulate various social platforms, including X and Reddit, and has been used to replicate key social phenomena such as information diffusion, polarization, and herd behavior. In our experiments, we use the population simulation component of OASIS. Since there is no official release of the OASIS profiles database, we follow the original methodology to generate personas based on the available demographic information in the Big-Five dataset. Specifically, we take subject profiles from the Big-Five dataset and apply OASIS’s simulation method to generate a descriptive summary for each individual. These summaries are inserted in a template as how OASIS does as system prompts. As the MobLab dataset does not contain demographic data, we exclude it from this evaluation. We uniformly sample 1,000 subjects without replacement and test the resulting simulation result using the GPT-4o model.

Modular Pluralism [1] is a multi-LLM framework that connects a base model (GPT-4o) with 11 smaller community language models representing diverse cultural (across five continents) and political (left, center, right) perspectives. Each community LM is initialized from Mistral-7B Instruct-v0.2 [3] and further finetuned on community-specific corpora using LoRA [2] for parameter-efficient adaptation. The original authors provide all community models, and the method is compatible with black-box LLMs without requiring retraining. We follow the original implementation for both model construction and usage.

1,000 Human Samples serves as an upper-bound reference constructed from real human data. We uniformly sample 1,000 responses without replacement from the dataset’s human annotations and use them to compare against the distributions generated by other models. This provides a grounded benchmark for evaluating the alignment and diversity of model outputs.

References

- [1] Shangbin Feng, Taylor Sorensen, Yuhao Liu, Jillian Fisher, Chan Young Park, Yejin Choi, and Yulia Tsvetkov. Modular pluralism: Pluralistic alignment via multi-llm collaboration. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 4151–4171, 2024.
- [2] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3, 2022.
- [3] Fengqing Jiang. Identifying and mitigating vulnerabilities in llm-integrated applications. Master’s thesis, University of Washington, 2024.
- [4] Yutong Xie, Zhuoheng Li, Xiyuan Wang, Yijun Pan, Qijia Liu, Xingzhi Cui, Kuang-Yu Lo, Ruoyi Gao, Xingjian Zhang, Jin Huang, et al. Be. fm: Open foundation models for human behavior. *arXiv preprint arXiv:2505.23058*, 2025.
- [5] Shaoyang Xu, Yongqi Leng, Linhao Yu, and Deyi Xiong. Self-pluralising culture alignment for large language models. *arXiv preprint arXiv:2410.12971*, 2024.
- [6] Ziyi Yang, Zaibin Zhang, Zirui Zheng, Yuxian Jiang, Ziyue Gan, Zhiyu Wang, Zijian Ling, Jinsong Chen, Martz Ma, Bowen Dong, Prateek Gupta, Shuyue Hu, Zhenfei Yin, Guohao Li, Xu Jia, Lijun Wang, Bernard Ghanem, Huchuan Lu, Chaochao Lu, Wanli Ouyang, Yu Qiao, Philip Torr, and Jing Shao. Oasis: Open agent social interaction simulations with one million agents, 2024. URL <https://arxiv.org/abs/2411.11581>.
- [7] Yaowei Zheng, Richong Zhang, Junhao Zhang, Yanhan Ye, Zheyang Luo, Zhangchi Feng, and Yongqiang Ma. Llamafactory: Unified efficient fine-tuning of 100+ language models. *arXiv preprint arXiv:2403.13372*, 2024.